KE-45

One-component Room Temperature Curing Type Liquid Silicone Rubber

1 Features and Benefits

1) One-component ready to use
2) Room temperature curing type (Condensation reaction oxime type)
3) Excellent adhesion to resin, glass and metal, etc.
4) Excellent heat and cold resistance.
   Continuous usable temperature is -40°C to 180°C.
5) Excellent electrical properties
6) Excellent weather resistance and durability

2 Applications

General industrial bonding and sealing

3 General Properties

Parameter \ Product name		KE-45
Curing method		One-component room temperature curing
Reaction mechanism (by-product gas)		Condensation(oxime)
Before curing		
Consistency		Paste
Tack-free time	min	6
Recommended curing conditions		23±2℃/50±5%RH×7days
After curing		
Density	g/cm^3	1.04
Hardness Durometer A		30
Tensile strength	MPa	2.0
Elongation at break	%	350
Volume resistivity	TΩ・m	5.0
Dielectric breakdown strength※	kV/mm	23
Relative permittivity 50Hz		3.0
Dielectric dissipation factor 50Hz		5×10^{-3}
Thermal conductivity	W/m・K	0.21
Tensile lap-shear strength	MPa	1.0 (Aluminum)

※Measured by 1mm thickness. (Not specified values)

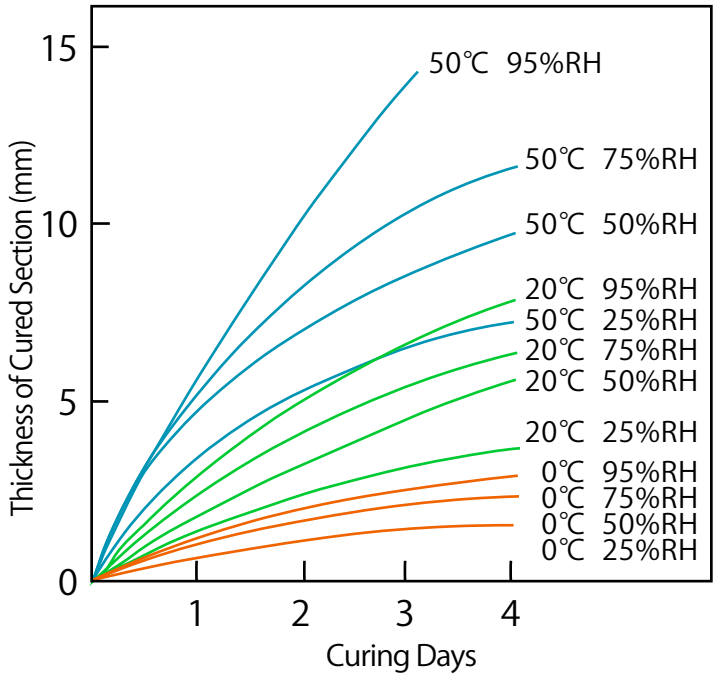
4 Packaging and Color

Packaging	100g (tube)				
Color	White	Translucent	Black	Reddish brown	Ivory
Product name	KE-45-W	KE-45-T	KE-45-B	KE-45-R	KE-45-YW

Packaging	330mL (cartridge)					
Color	White	Translucent	Black	Gray	Reddish brown	Ivory
Product name	KE-45-W	KE-45-T	KE-45-B	KE-45-G	KE-45-R	KE-45-YW

5 Curing Properties

■ KE-45 (Condensation reaction oxime type)
Relationship between curing speed and temperature and humidity



6 Weather Resistance and Durability

■ KE-45 (Condensation reaction oxime type) - Results of outdoor exposure testing

Physical properties of rubber

Parameter / Exposure period	Hardness Durometer A	Tensile strength MPa	Elongation at break %	Estimated luminous intensity J/m²			Estimated precipitation mm
				Ultraviolet rays	Visible light rays	Infrared rays	
Initial	30	2.3	350	—	—	—	—
1 month	35	2.0	370	1.60×10^{7}	6.44×10^{7}	9.13×10^{7}	21
3 months	34	2.0	330	5.46×10^{7}	2.81×10^{8}	3.00×10^{8}	63
6 months	37	2.0	360	1.44×10^{8}	7.74×10^{8}	8.80×10^{8}	335
1 year	37	2.0	320	3.00×10^{8}	1.63×10^{9}	1.59×10^{9}	1,376
2 years	37	1.8	310	5.87×10^{8}	3.33×10^{9}	3.32×10^{9}	2,130

Testing method : Complies with JIS K 6249 (Not specified values)
※ The PH-11M-2AT actinometer was used in the tests.

Adhesion

Parameter / Exposure period	Maximum tensile stress N / mm²	Cohesion break rate %	Estimated luminous intensity J/m²			Estimated precipitation mm
			Ultraviolet rays	Visible light rays	Infrared rays	
Initial	0.70	100	—	—	—	—
1 month	0.67	100	1.70×10^{7}	9.39×10^{7}	9.03×10^{7}	28
3 months	0.69	100	6.75×10^{7}	3.98×10^{8}	3.57×10^{8}	123
6 months	0.71	100	1.72×10^{8}	9.79×10^{8}	9.01×10^{8}	413
1 year	0.70	100	3.01×10^{8}	1.70×10^{9}	1.61×10^{9}	1,361
2 years	0.71	100	5.82×10^{8}	3.37×10^{9}	3.31×10^{9}	2,154

Substrate : Glass, PRIMER-C used. (Not specified values)
Testing method : complies with JIS A 1439
※ The PH-11M-2AT actinometer was used in the tests.

7 Handling Precautions

1) One-component condensation-cure liquid silicone rubbers cure by reacting with moisture in the air, and curing starts at the surface. Curing speed is affected by temperature, humidity and other conditions.
   These products do not have particularly good deep-section cure properties, and are thus not suitable for adhesive applictions in which the contact area is large. Additionally, if the product is used when humidity is above 100% and water droplets form on the rubber during the cure process, a hydrolysis reaction will proceed ahead of the crosslinking curing reaction, in which case the rubber may have reduced strength or exhibit surface tack after curing.
2) Oxime type condensation-cure liquid silicone rubber may corrode copper metals under sealed conditions. Please conduct a test using a small sample to determine whether the product is suitable for the intended applications.
3) Condensation-cure liquid silicone rubbers will show a temporary decline in dielectric properties during the cure process. In most cases, however, the rubber will recover and exhibit its intrinsic dielectric properties when fully cured.
4) Please note that in some cases, the rubber may not cure if it comes in contact with flux or certain other materials.
5) Condensation-cure liquid silicone rubber products should not be used in place where completely airtight conditions will be created.
6) Condensation-cure liquid silicone rubbers may yellow over time, but their other characteristics will not be affected.

8 Precautions when Using

1) Be sure to clean the substrate to remove dirt, grime, moisture and oil from the surface.
2) When using an air gun applicator, be sure to set the pressure at a safe and suitable level, around 0.2–0.3 MPa MAX.
3) Please contact your sales representative if you have any questions regarding the handling and use of these products.

9 Safety and Hygiene

1) Be sure there is adequate ventilation when using condensation-cure liquid silicone rubber products.
   This product generates methyl ethyl keton oxime (MEKO) when cured.
   If you experience unpleasant symptoms when using these products, move to an area with fresh air.
2) Uncured liquid silicone rubber products may irritate skin and mucous membranes.
   Take care to avoid eye contact or prolonged contact with the skin.
   In case of accidental eye contact, immediately flush with water for at least 15 minutes and then seek medical attention. In case of skin contact, wipe off immediately with a dry cloth and then wash thoroughly with soap and water. Contact lens wearers must take special care when using liquid silicone rubber : if uncured liquid silicone rubber enters the eye, the contact lens may become stuck to the eye.
3) Never touch or rub the eyes while working with these products. Users should wear safety glasses and take other appropriate steps to protect their safety.
4) If product gets on the floor, it will become slippery. Wipe the product to remove completely.
5) Keep out of reach of children.
6) Be sure to read the Safety Data Sheets (SDS) for these products before use.
   SDS are available from the Shin-Etsu Silicone website.
   If the SDS is not listed on the website, please contact the sales department.

   SDS download URL :
   https://www.shinetsusilicone-global.com/support/sdstds →

10 Storage Conditions

Storage Temperature : 1℃ to 30℃
※Seal container tightly and store in a dark place (out of direct sunlight) with good ventilation. Keep away from heat and flame.

11 UN Number

N / A

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Shin-Etsu Chemical Co.,Ltd.
Marunouchi Eiraku Bldg., 4-1, Marunouchi 1-chome, Chiyoda-ku, Tokyo 100-0005, Japan
Silicone Division Sales and Marketing Department IV
Phone : +81-(0)3-6812-2410 Fax : +81-(0)3-6812-2415

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